Cyber Threat Intelligence System

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***Abstract*—The rapid growth of cyber threats across publicly accessible online platforms has made traditional reactive security mechanisms insufficient for timely threat mitigation. This paper presents a scalable Cyber Threat Intelligence (CTI) system that integrates automated web crawling, natural language processing, and machine learning for real-time threat detection. The proposed architecture continuously monitors open web sources, including forums, news portals, and social media platforms, to collect and preprocess unstructured textual data. A rule-based keyword filtering mechanism reduces irrelevant content, after which a fine-tuned TinyBERT model performs multi-class threat severity classification. The system categorizes threats into distinct severity levels to support prioritized response and decision-making. Extracted intelligence is stored in a structured database and visualized through a dashboard interface that enables monitoring, analytics, and alert generation. Experimental evaluation demonstrates strong classification performance, with the proposed model outperforming conventional machine learning and deep learning baselines in accuracy, precision, recall, and F1-score. The overall pipeline provides an efficient and automated solution for proactive cyber threat monitoring, helping organizations strengthen situational awareness and enhance incident response capabilities.**

***Index Terms*—Cybersecurity, Web Crawler, TinyBERT, NLP, Threat Detection, MongoDB**

I. Introduction

The Internet is a vast and constantly changing environment in which cybercriminals take advantage of loopholes to carry out phishing attacks, distribute malware, and disclose confidential information. Threat actors leverage publicly accessible open web platforms, including social media, online forums, and online retailers, to converse, plan attacks, and disseminate malicious content. As cyberspace threats continue to evolve, organizations find themselves facing ever-difficult times detecting and countering risks that originate from such open web sources. The objective of this project is to monitor and examine online activities in real time, thus facilitating early threat detection and actionable intelligence to organizations. With the use of sophisticated web scraping methods, machine learning patterns, and real-time data analysis, the system will detect emerging threats, malicious activity, and potential cybersecurity loopholes before they gain momentum. To ensure scalability, security, and efficiency, the system is deployed on a cloud-based infrastructure that supports high-performance computation, secure data management, and seamless integration with existing cybersecurity systems. This pre-emptive strategy allows organizations to remain ahead of emerging cyber threats, enhance their defense measures, and respond quickly to potential risks, thus enhancing their overall security posture. Conventional security devices are severely hindered in cyber threat monitoring due to the unstructured and fragmented nature of online information. Cyber threats are likely to be launched, malware propagated, and vulnerabilities discussed using forums, social networking sites, and messaging applications by malicious players. Some of the primary issues include the fragmentation of threat intelligence across multiple platforms, making it more difficult to gather and analyze corresponding intelligence; the absence of real-time monitoring capabilities, which causes slower reactions to evolving cyber threats, as the large volume and complexity of data make manual analysis impractical and resource-intensive; and the inability to identify emerging threats, as the methods used by cybercriminals change rapidly and are outside the capabilities of conventional security devices. This creates the need for a scalable, real-time automated monitoring mechanism capable of continuously monitoring online activities and delivering timely, actionable intelligence to facilitate effective threat identification and response.

II. Literature Review

Cyber Threat Intelligence (CTI) has evolved as a critical component in modern cybersecurity frameworks. Early discussions on cybersecurity challenges emphasized the growing complexity of threats and the need for adaptive defense mechanisms [1]. In IoT and cyber-physical systems, security and forensic challenges were comprehensively analyzed in [2], highlighting inherent vulnerabilities and the necessity of structured investigative methodologies. Similarly, data-driven CTI for real-time attack detection in cyber-physical systems was proposed in [3], demonstrating the importance of analytics-driven security architectures.

Structured CTI integrated with machine learning has gained significant attention. Ahmadzadeh et al. [4] introduced a machine learning-based framework leveraging structured threat intelligence to enhance detection accuracy. A broader survey of CTI research trends and future directions was provided in [5], identifying gaps in automation, intelligence sharing, and scalability. Standardization and taxonomy-related challenges within CTI ecosystems were further evaluated in [6], while threat actor profiling and characterization techniques were explored in [7].

Dark web and deep web intelligence extraction has emerged as a proactive defense strategy. Chen et al. [8] demonstrated methodologies for mining dark web data to uncover emerging cyber threats. Complementary approaches for darknet intelligence collection were proposed in [9], while proactive exploit detection using online vulnerability discussions was explored in [10]. Neural language modeling for exploit prediction was introduced through DarkEmbed in [11], showcasing the effectiveness of deep learning for anticipating cyber attacks.

Anomaly detection and machine learning-based intrusion detection systems have been widely investigated. A comprehensive review of anomaly detection methods in cybersecurity was presented in [12]. Ensemble-based feature selection techniques for DDoS detection in cloud environments were proposed in [13]. Advanced intrusion detection models for IoT networks using dimension reduction and multi-tier classification were introduced in [14]. Scalable unsupervised deep learning systems for smart grid attack detection were developed in [15], while machine learning-assisted malware classification for Android platforms was investigated in [16].

Recent advancements integrate artificial intelligence, knowledge graphs, and large language models into CTI pipelines. An AI-enhanced CTI processing pipeline was proposed in [17], focusing on automation and intelligent threat analysis. The use of knowledge graphs combined with large language models for generating actionable CTI was explored in [18], demonstrating improved contextual reasoning capabilities. Additionally, information propagation dynamics relevant to influence and threat spread modeling were examined in [19], providing insights into adversarial behavior dissemination patterns.

Overall, existing literature highlights significant progress in CTI automation, dark web intelligence mining, and machine learning-driven detection systems. However, challenges remain in integrating structured intelligence, scalability, and real-time actionable threat response, motivating further research in intelligent and unified CTI frameworks.

III. Proposed Method

The proposed Cyber Threat Intelligence System functions as a multi-stage pipeline, consisting of web crawling, text processing, keyword-based violation tagging, TinyBERT-based severity classification, and visual analytics for actionable threat monitoring. Architecture and internal workflow are presented in Fig. 1.

A. Web Crawling and Scraping

A custom web crawler periodically scans surface-level websites of forums, marketplaces, blogs, and news portals. The approach uses HTTP requests to fetch HTML pages, scrape them, and store them in text form.

B. Data Preprocessing and Cleaning

HTML content extracted gets parsed through BeautifulSoup, where non-text elements, scripts, HTML tags, and advertising fragments get removed. Regular expressions normalize whitespace and strip unwanted characters. After preprocessing, the clean corpus is forwarded to the classification modules.

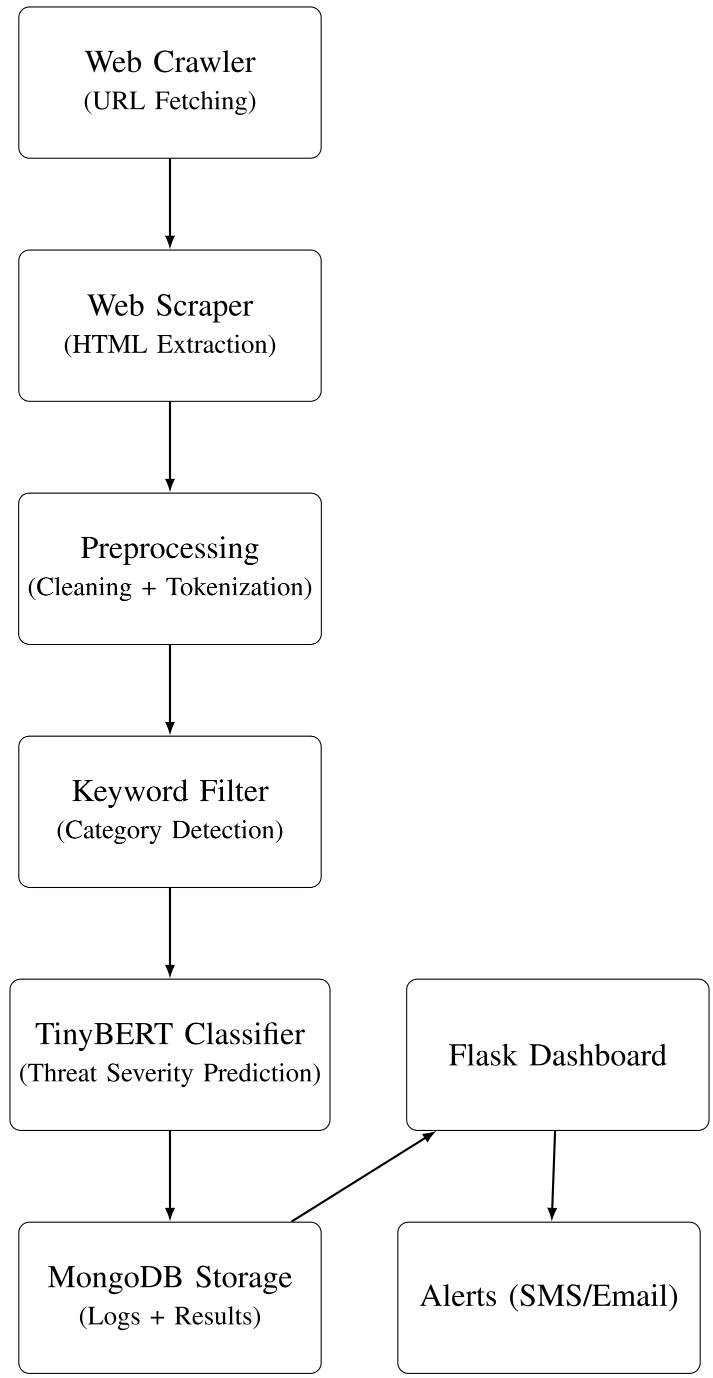

Fig. 1. System Architecture Diagram

C. Rule-Based Keyword Violation Detection

A keyword filtering unit identifies category-specific terms associated with threat domains relating to the areas of hacking, malware, fraud, weapons, and narcotics. If any category keyword k_i is detected in text segment S, the tagging function is :

$$\textbf{Category}(S) = \begin{cases} C_i, & \text{if } k_i \in S \\ \text{None}, & \text{otherwise} \end{cases}$$

This stage reduces model load by filtering irrelevant pages before TinyBERT inference.

D. TinyBERT-Based Threat Severity Classification

The processed text is fed into a fine-tuned TinyBERT transformer, which embeds, encodes, and assesses the threat intensity of the input tokens with a maximum length of 512. Severity is categorized in four levels: None, Low, Medium, and High.

E. Database and Dashboard Integration

Classified results (URL, extracted text, category, severity, and timestamp) are stored in MongoDB. A Flask dashboard fetches records to be visualized via charts, filters, and highlighting based on severity. If Severity $\geq$ High, alert triggers are automatically generated via mail/SMS.

F. Complete Pipeline Execution

1) Crawler fetches new URLs periodically
2) Scraper extracts and formats web content
3) Preprocessor filters and normalizes text
4) TinyBERT predicts severity and category
5) Database stores structured threat intelligence
6) Dashboard visualizes metrics and sends live alerts

This combined methodology enables real-time scalable web threat monitoring using NLP-driven classification supported by quantitative evaluation.

IV. Results

This section provides a performance evaluation of the Cyber Threat Intelligence System that integrates a web crawler and a TinyBERT-based text classification model. The results cover crawler efficiency metrics, TinyBERT model evaluation metrics, and their comparison to existing baseline models. All experiments were conducted using the collected dataset of scraped web pages after having been preprocessed and normalized.

A. Crawler Performance

It was considered necessary to evaluate the custom crawler based on its efficiency in text extraction, cleaning, and processing from various online sources. The metrics include crawling speed, quality of extracted content, efficiency of noise removal, and preprocessing time. Table I summarizes the crawler's performance. Mathematically, for measurement of system efficiency, let crawling performance be defined as:

$$\textbf{Crawling Speed} = \frac{P}{T} \quad \text{(pages/sec)}$$

$$\textbf{Data Extraction Efficiency} = \frac{\text{Relevant Text Extracted}}{\text{Total Raw Data Collected}} \times 100$$

where P is total pages crawled and T is total processing time.

TABLE I
CRAWLER PERFORMANCE METRICS

Metric	Value
Pages Crawled per Minute	45
Total Pages Processed	9,560
Crawl Success Rate (%)	90.4
Average Response Time (ms)	369
Average Content Size (KB)	46.8
Noise Removed (%)	80.1
Preprocessing Time per Page (ms)	198

The crawler performance trend over different intervals is illustrated in Fig. 2.

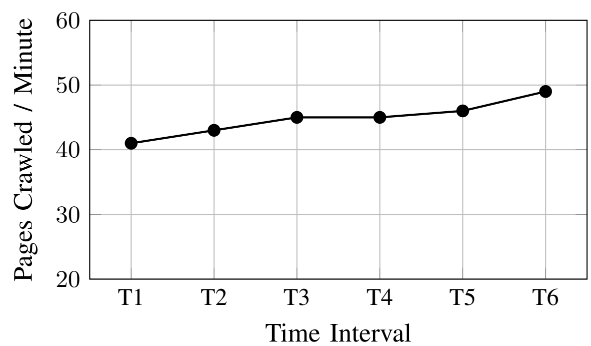

Fig. 2. Crawler Performance Over Time

B. TinyBERT Model Evaluation

Metrics such as accuracy, precision, recall, F1-score, AUC, and inference time are considered among the standard machine learning metrics that were considered for evaluating the TinyBERT classifier. These give insight into the model's effectiveness in threat classification.

To quantify model accuracy using prediction outputs:

$$\textbf{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\textbf{Precision} = \frac{TP}{TP + FP}$$

$$\textbf{Recall} = \frac{TP}{TP + FN}$$

$$\textbf{F1-Score} = 2 \times \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

For multi-class severity evaluation, weighted averaging is used:

$$\textbf{Weighted F1} = \sum_{i=1}^{n} w_i \times F1_i, \quad w_i = \frac{N_i}{N_{\text{total}}}$$

Table II presents the detailed results.

TABLE II
TINYBERT MODEL PERFORMANCE METRICS

Metric	Value
Accuracy (%)	90.2
Precision (%)	90.6
Recall (%)	90.8
F1-Score (%)	89.2
AUC–ROC	0.924
Training Time per Epoch (s)	98
Inference Time per Sample (ms)	18.4

The training and validation accuracy progression is shown in Fig. 3.

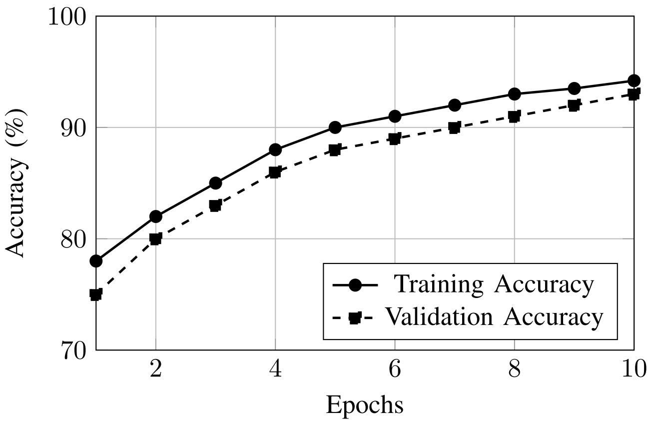

Fig. 3. Training and Validation Accuracy of TinyBERT Model

The impact of training data size on classification accuracy is presented in Fig. 4.

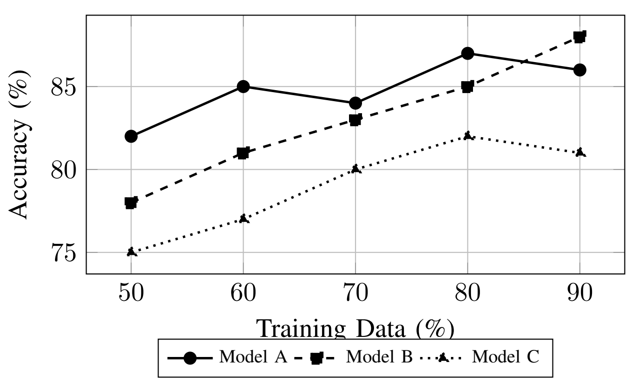

Fig. 4. Accuracy vs Training Data Percentage

Precision performance across varying training data sizes is illustrated in Fig. 5.

The variation of F1-score with training data percentage is shown in Fig. 6.

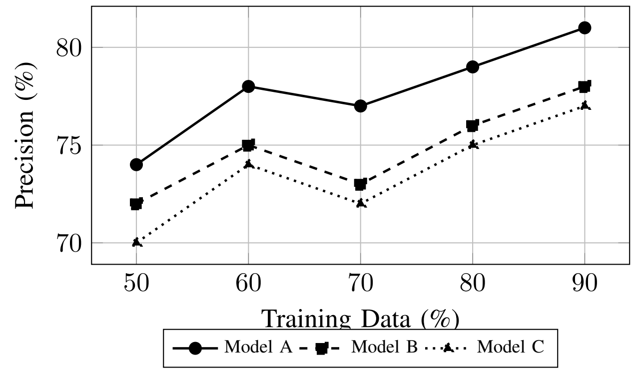

Fig. 5. Precision vs Training Data Percentage

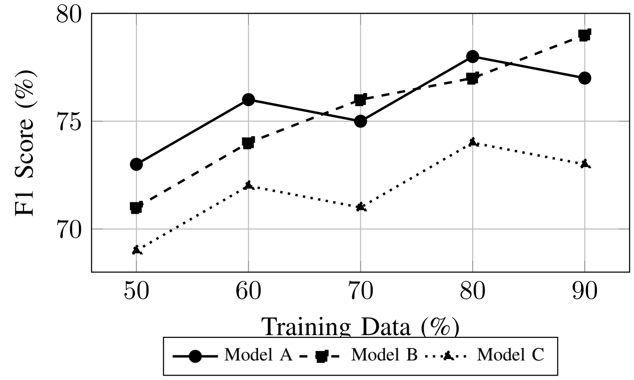

Fig. 6. F1-Score vs Training Data Percentage

C. Comparison With Existing Methods

To evaluate the effectiveness of the proposed architecture, the TinyBERT model was compared with conventional machine learning and deep learning baselines, including Naive Bayes, Logistic Regression, Random Forest, and LSTM. A detailed comparison of performance metrics—accuracy, precision, recall, F1-score, and inference time—is presented in Table III.

TABLE III
COMPARISON OF PROPOSED TINYBERT MODEL WITH EXISTING METHODS

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-S
Naive Bayes	80.3	78.6	78.2	
Logistic Regression	84.5	83.1	82.3	
Random Forest	85.2	84.4	83.1	
LSTM	86.7	85.8	84.9	
Proposed TinyBERT Model	**90.2**	**90.6**	**90.8**	

A metric-wise visual comparison of models is provided in Fig. 7.

D. Discussion

Crawler results show that the system can efficiently crawl and preprocess web data with a high success rate and significant noise reduction. The TinyBERT model demonstrates strong classification performance, outperforming all baseline

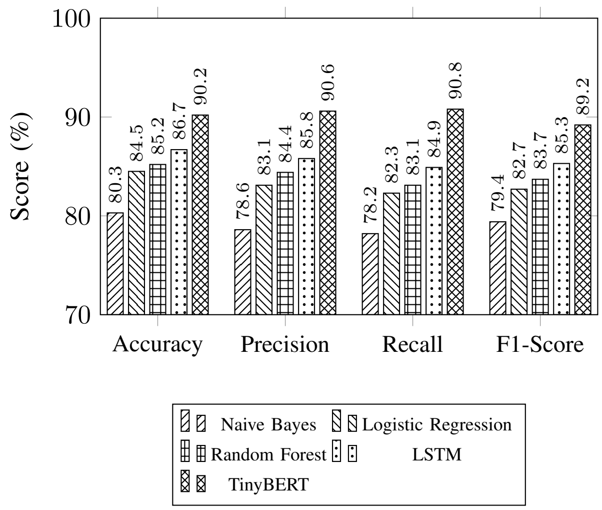

Fig. 7. Compact Metric-wise Performance Comparison

models across key evaluation metrics. The combined pipeline is a robust and scalable end-to-end threat intelligence solution capable of processing volumes of online content in near real time.

V. Conclusion and Future Work

In the evolving cybersecurity landscape, traditional security mechanisms are increasingly insufficient to address emerging and sophisticated cyber threats. Organizations need to be proactive about the risks that come with publicly accessible platforms. This project introduces an automated threat intelligence system that uses web scraping to keep an eye on online activities in real time. By harnessing cutting-edge techniques like natural language processing and machine learning, the system delivers timely and actionable insights into potential cyber threats. With its scalable and secure deployment on the Cloud, it ensures high-performance data processing and fits seamlessly into existing security frameworks. Ultimately, this solution empowers organizations to stay one step ahead of cybercriminals, respond quickly to threats, and bolster their overall security posture.

While the current system shows great promise in detecting threats early, there are several ways to enhance its effectiveness even further. Expanding its reach to include dark web forums and marketplaces would provide a deeper understanding of potential threats, and adding multilingual NLP models would help identify risks from various linguistic backgrounds. Developing a threat prioritization engine could rank threats based on their severity, source credibility, and potential impact, allowing for smarter triage. Moreover, an advanced visualization dashboard could present interactive, real-time insights into threat patterns and trends. Integrating behavioral analysis could also help link external threats with internal vulnerabilities, giving a more comprehensive view of risk. These future directions aim to transform the system into a robust, adaptive cybersecurity intelligence solution that can tackle even the most sophisticated and emerging threats.

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